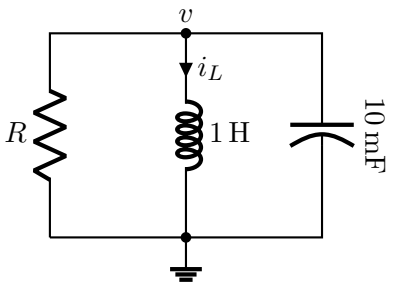
EE2015: Electric Circuits and Networks

Tutorial 7

(October 3, 2025)

1. For the circuit shown on the right it is given that $v(0) = 5\,V$ and $i_L(0^-) = 0$. Find $v(t)$ for $t \geq 0$ when (i) $R = 1.923\,\Omega$, (ii) $R = 5\,\Omega$, and (iii) $R = 6.25\,\Omega$.

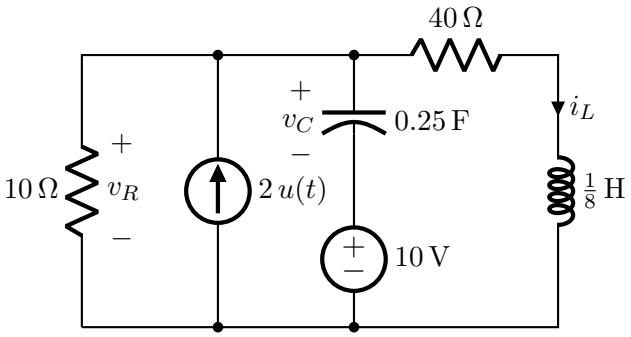


2. The responses of a series RLC circuit are, for $t \geq 0$, given by

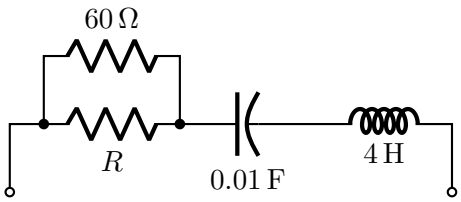
$$v_C(t) = 30 - 10\,e^{-20t} + 30\,e^{-10t}\,\text{V}$$
$$i_L(t) = 40\,e^{-20t} - 60\,e^{-10t}\,\text{mA}$$

Determine the values of R, L, and C that are consistent with the given expressions.

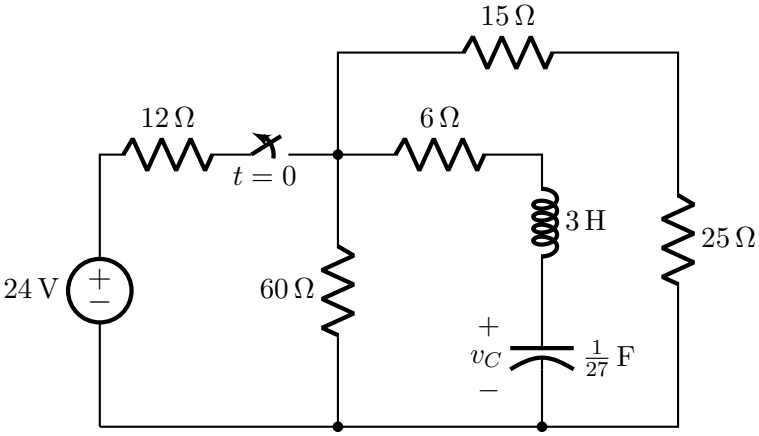
3. For the circuit shown on the right find the following:
   (a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$
   (b) $i_L'(0^+)$, $v_C'(0^+)$, $v_R'(0^+)$
   (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$



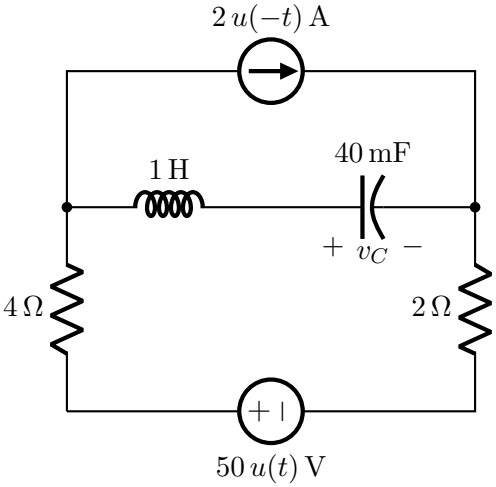
4. For the circuit shown on the left, find the value of R such that will give rise to a crtically damped response.



5. For the circuit shown on the right find $v_C(t)$ for $t \geq 0$. Assume that the switch has been in the closed position for a long time prior to $t = 0$.

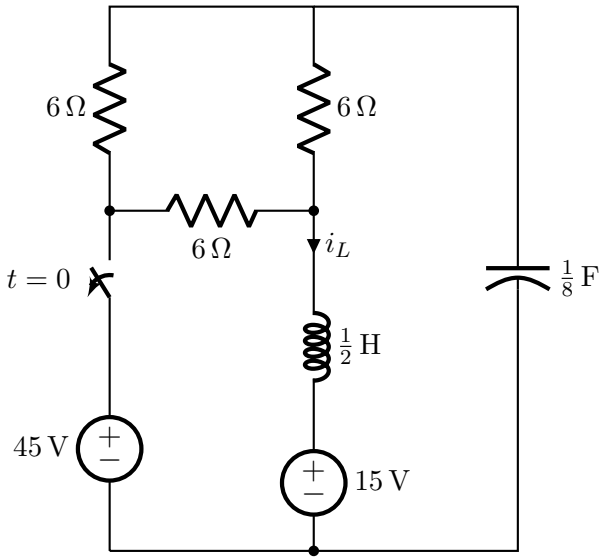


6.



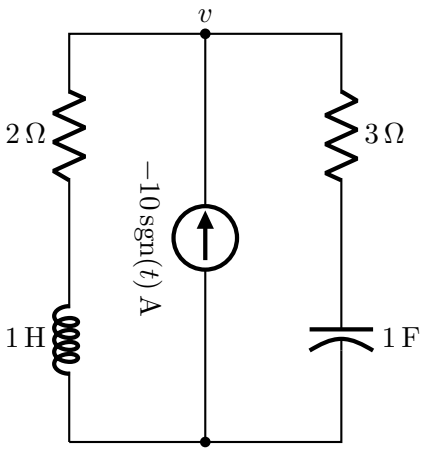
For the circuit shown on the right find $v_C(t)$ for $t \geq 0$.

7.



For the circuit shown on the left, find $i_L(t)$ for $t \geq 0$.

8.



Find $v(t)$ for $t \geq 0$.